```
For each supplier, calculate:

Number of products supplied
Average MRP of those products
Maximum MRP of those products

Display only suppliers who supply at least 3 products.
Sort them by the number of products supplied from highest to
lowest.



select
  SupplierID,
  count(*) as no_of_products,
  avg(MRP) as avgMRP,
  max(MRP) as maximumMRP
from `e1.products`
group by SupplierID
having no_of_products >= 3
order by no_of_products desc


Find each CategoryID where the average MRP of products is
greater than ₹15,000.

Display:

CategoryID
Number of products
Average MRP
```

```
Maximum MRP

Sort categories by average MRP from highest to lowest.



select
  CategoryID,
  count(*) as no_of_products,
  avg(MRP) as avg_mrp ,
  max(MRP) as max_mrp
from `e1.products`
group by CategoryID
having avg_mrp > 150
order by avg_mrp desc



Using order_items, calculate the total quantity sold and
total sales value for every product.

Display only products where:

Total quantity sold is at least 10
Total sales value is greater than ₹1,00,000

Sort by total sales value descending.



select
  ProductID,
  sum(Quantity) as total_quantity,
```

```
  sum(total) as total_price,
from `e1.order_items`
group by ProductID
having
  total_quantity >= 10 and
  total_price > 100000
order by total_price desc
```

Find products where the selling price is lower than the MRP.

Display:

ProductID
ProductName
MRP
SellingPrice
Discount amount

Sort products by the highest discount amount first.

```
select
  ProductID,
  ProductName,
  MRP,
  SellingPrice,
  (MRP - SellingPrice) AS discount_amount
from `e1.products`
where SellingPrice < MRP
order by discount_amount desc
```

```
select *
from `e1.products`
limit 5


P1 --> 2k discount
P2 --> 1K discount

100000 --> 98000 --> 2% disc
2000 --> 1000 --> 50% disc




Find products where the discount is at least 10% of MRP.

Display:

ProductID
ProductName
MRP
SellingPrice
Discount percentage

Sort by discount percentage from highest to lowest.


select
```

```
  ProductID,
  ProductName,
  MRP,
  SellingPrice,
  round(((MRP-SellingPrice)/MRP),2)*100 as discount_percentage
from `e1.products`
where round(((MRP-SellingPrice)/MRP),2)*100 >= 10
order by discount_percentage desc
```

Consider only successful payments.

For each payment method, calculate:

Number of successful transactions
Total payment amount
Average payment amount
Maximum payment amount

Display only methods where the total successful payment amount exceeds ₹1,00,000.

```
select
  Method,
  count(*) as no_of_sucessful_transaction,
  sum(Amount) as total_amount,
  round(avg(Amount),2) as avg_amount,
```

```
  max(Amount) as mx_amount
from `e1.payments`
where Status="Success"
group by Method
having total_amount>100000;
```

Find customers who have placed between 4 and 8 orders, inclusive.

Display:

CustomerID
Number of orders

Sort customers by order count descending and display only the top 10.

```
select
  CustomerID,
  count(*) as num_of_orders
from `e1.orders`
group by CustomerID
having num_of_orders between 4 and 8
order by num_of_orders desc
limit 10
```

For each warehouse, calculate:

Total stock
Average stock
Highest stock for a product
Lowest stock for a product

Display only warehouses where total stock is greater than 500.

Sort by total stock descending.

```
select
  WarehouseID,
  sum(Stock) as total_stock,
  round(avg(Stock),2) as avg_stock,
  max(Stock) as highest_stock,
  min(Stock) as lowest_stock
from `e1.inventory`
group by WarehouseID
having total_stock > 500
order by total_stock desc
```

Find products that have at least one inventory record where stock is below 20 units.

Display:

ProductID
Number of inventory records
Minimum stock
Maximum stock

Show only products whose minimum stock is below 20.

```
select *
from `e1.inventory`
limit 5
```

```
select
  ProductID,
  count(*) as no_of_rec,
  max(Stock) as max_stk,
  min(Stock) as min_stk
from `e1.inventory`
group by ProductID
having min_stk < 20
order by min_stk
```

DPP Questions:

Calculate the number of orders and percentage of total orders represented by each order status.

Display the most common status first.

Using order_items, find the top 5 products based on total quantity sold.

Display:

ProductID
Total quantity sold
Number of order-item records
Total sales value

Consider only order-item records where the total value of that item is greater than ₹25,000.

For each product, calculate:

Number of such records
Total quantity
Total sales value

Display products having at least 3 qualifying records.

For each product, calculate:

Minimum selling price
Maximum selling price
Average selling price
Number of order-item records

Display only products where the minimum selling price is greater than ₹10,000.

Sort by average selling price descending.

Calculate the total sales value generated by each supplier.

You will need to connect the products and order_items tables.

Display:

SupplierID
Number of products supplied
Total sales value

```
Sort suppliers by total sales value descending.
```